Sparse Information Perception Network for Remote Sensing Cross-Modal Retrieval

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***Abstract*—In recent years, research on remote sensing (RS) cross-modal retrieval (RSCMR) has gained significant attention, yet the inherent sparsity of RS images has posed challenges in aligning them with corresponding text, leading to reduced retrieval performance. To address this, we propose the sparse information perception network (SIPN), an instance-level dynamic network that adapts to sparse information in RS images and text. Our approach incorporates a dynamic feature fusion module (DFFM) for initializing the network, dynamic training adjustments using random processes and pretrained policy networks, and feature alignment via contrastive loss. Additionally, we enhance feature selection and fusion capabilities through a multilevel attention module (MLAM), which includes cross-attention (CA) and a multilayer perceptron (MLP), with triplet loss for further alignment. Our method demonstrates superior performance in RSCMR on the RSICD and RSITMD datasets.**

***Index Terms*—Dynamic network, information fusion, remote sensing (RS) cross-modal text–image retrieval.**

I. Introduction

With the advancement of Earth observation technologies, collecting vast amounts of remote sensing (RS) images is no longer a challenging task [1], [2], [3]. This has provided new opportunities for scholars to observe the Earth. The need to extract information content from these RS images has given rise to various downstream tasks [4]. From the perspective of RS image retrieval, there are primarily two approaches: content-based retrieval [5], [6], [7] and cross-modal retrieval [8], [9], [10]. The former involves retrieving images similar to a query RS image based on content similarity, emphasizing the understanding of image content. The latter focuses on retrieving different modalities of data, usually text, that share similar semantics with RS images. This task emphasizes establishing connections between two different modalities of data: images and text. RS cross-modal retrieval (RSCMR) bridges the gap between image understanding and language comprehension. It has gained increasing attention in society and found numerous practical applications [11], [12], [13].

Research on RSCMR has been ongoing for some time, but it is still in the exploratory phase. With the development of deep learning technologies, multiple RSCMR [9], [10], [14] methods have been proposed. These methods often share a similar dual-branch structure: one branch extracts visual features, while the other branch extracts text features. The features obtained from these two branches often have different dimensions, so they need to be mapped to a common space through a linear layer and then using a loss function. Typically, the visual feature extractor is a convolutional neural network (CNN) [15], while the text feature extractor is a recurrent neural network (RNN) [16]. The loss for aligning features can be selected from triplet loss [17], contrastive loss [18], or a combination of both.

However, existing methods mainly focus on image–text matching, without adequately considering the unique characteristics of RS images. In contrast to traditional cross-modal retrieval tasks, RS retrieval presents unique challenges due to the presence of sparse visual information in certain types of images. The sparsity of RS images refers to the limited diversity and low entropy of their inherent visual features. As illustrated in Fig. 1, certain land-cover types—such as forests, barelands, ponds, and playgrounds—exhibit an inherent tendency toward information sparsity in RS imagery. For instance, barelands typically contain minimal visual diversity due to homogeneous soil color and texture, resulting in low entropy across the image. Forest scenes, despite their apparent visual complexity, often present sparse information when the canopy is uniform and dominated by repetitive textures, with meaningful variation primarily occurring along edges. Similarly, pond surfaces generally display consistent coloration and smooth reflections, while playgrounds often consist of flat, featureless surfaces lacking distinctive structures or activities. Specifically, image categories such as bareland, forest, pond, and playground often exhibit large homogeneous regions in terms of color, texture, and structure, resulting in significantly reduced information density. This sparsity leads to partial mismatches between images and textual descriptions, which, in turn, often degrades retrieval accuracy [19], [20]. These conditions lead to intrinsically limited feature diversity, making such scenes prone to sparse representation. Traditional cross-modal retrieval methods rely heavily on object detection to establish semantic correspondences between images and textual descriptions. While effective for images rich in

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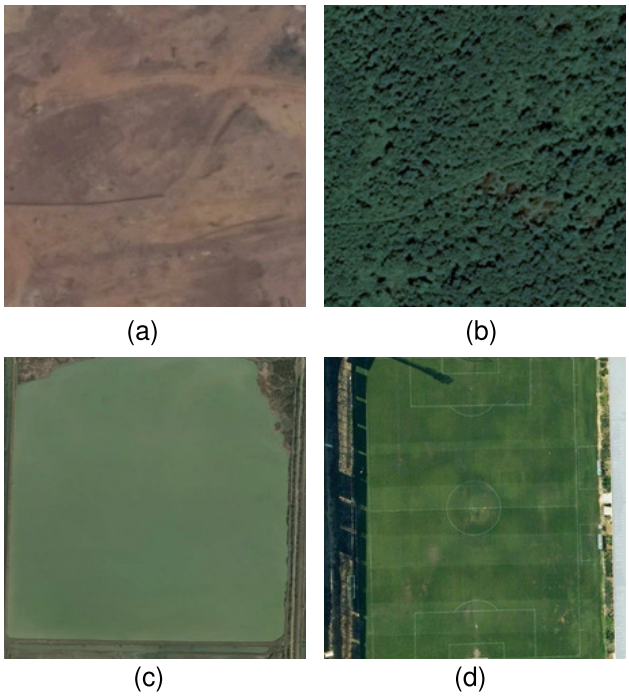


Fig. 1. Illustration of RS images with sparse information. Samples including (a) bareland, (b) forest, (c) pond, and (d) playground demonstrate characteristic sparsity, where textual descriptions provide minimal coverage or omit key visual elements. These categories often exhibit extensive homogeneous regions in color, texture, and structure, resulting in low information density. Such visual redundancy diminishes feature discriminability, hindering accurate text–image alignment. Addressing this sparsity is critical for enhancing cross-modal retrieval robustness and semantic fidelity in RS applications.

salient objects and structural detail, this paradigm becomes inadequate for RS images where prominent objects are often absent or indistinct. To overcome this limitation, we propose an instance-level dynamic network specifically designed to address the information-sparse characteristics of RS data. By focusing on adaptive feature extraction and alignment, our approach enables robust image–text matching even in scenarios with limited perceptual cues.

To address the aforementioned issue, we propose a sparse information perception network (SIPN). The SIPN comprises several essential modules, each meticulously designed to tackle information sparsity challenges and enhance cross-modal comprehension. These modules consist of the following.

1) *Dynamic Feature Fusion Module (DFFM):* The DFFM utilizes a stochastic process to stack feature fusion layers within the model dynamically. Traditional static network architectures' rigid feature fusion mechanism faces dual limitations when dealing with multimodal data heterogeneity and differences in input information density. To address this, the proposed dynamic fusion approach enhances adaptability across diverse data by ensuring improved visual representations for images with varying information levels. Additionally, the uncertainty introduced by the random process strengthens the model's ability to handle low-information-density images, ultimately improving subsequent perception probability.
2) *Prior-Aware Perception Module (PAPM):* The PAPM is a pretrained *policy network*. Recognizing that dynamic fusion through a stochastic process can introduce increased uncertainty, including noise and redundant information, the PAPM incorporates prior knowledge from cross-modal retrieval. This prior knowledge helps determine when to stop stacking feature fusion layers, thereby reducing the increase in uncertainty.
3) *Multilevel Attention Module (MLAM):* The MLAM is carefully designed to improve the alignment between images and text. Motivated by the information bottleneck theory, our approach advocates preserving task-relevant information while eliminating redundancy in feature representations, thereby reducing overfitting and alignment errors, especially under sparse or noisy data conditions. To further address the semantic imbalance across modalities—such as the contrast between background noise and salient regions in images, or between filler and key tokens in text—we propose a multilayer attention mechanism that dynamically reweights input components based on their informativeness. This enables the model to focus on semantically rich regions while mitigating the impact of less relevant content during feature alignment. As a result, our module effectively captures and aligns features across multiple semantic levels and granularities—including low-level visual patterns and high-level conceptual cues in images, as well as local lexical information and global contextual semantics in text—thereby facilitating more accurate and robust cross-modal semantic matching under challenging conditions. Accordingly, this module is optimized to extract features of various dimensions for cross-modal alignment. Extensive experiments are conducted on the RSICD [21] and RSITMD [10] datasets. The experimental results demonstrate the effectiveness of the proposed methods.

The main contributions of our work are as follows.

1) An SIPN is proposed to address the challenges of sparse information in RS cross-modal retrieval.
2) A dynamic network is introduced, leveraging random processes to augment the uncertainty in fusion. This enhancement is aimed at bolstering the model's capability to discern images characterized by low information density.
3) To enhance the model's ability to select and fuse features, a multilevel attention mechanism is introduced. Cross-attention (CA) is used to blend features, and a multilayer perceptron (MLP) is employed to reduce the dimensionality of long features, resulting in better representation.
4) The effectiveness of the proposed method is verified through experiments conducted on the RSICD and RSITMD datasets. Additionally, ablation studies are performed to demonstrate the effectiveness of each component.

II. Related Work

This section begins by providing an overview of the progress made in the field of RSCMR. It then delves into

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the utilization of information fusion techniques in the context of cross-modal retrieval.

A. RS Cross-Modal Retrieval

RSCMR is a field that focuses on the retrieval of relevant information from remote sensing images based on textual queries. Over the years, there have been significant developments in RSITR methods, aiming to improve retrieval accuracy and efficiency.

Most existing methods are embedded-based. The embedded-based RSCMR methods refer to mapping the RS image and text into the same high-dimensional space and measuring the cross-modal similarity by an appropriate distance. Abdullah et al. [14] introduced a deep bidirectional triplet network, leverages both textual and visual information to learn a shared embedding space, enabling effective cross-modal retrieval in the context of RS. Considering the characteristics of RS image information redundancy, Yuan et al. [10] employed the multiscale visual self-attention (MVSA) module to extract the salient features of the RS image and utilized visual features to guide the text representation. Cheng et al. [9] leveraged deep learning techniques and a novel semantic alignment loss that learns to align and match visual and textual representations, facilitating effective retrieval of RS images based on textual queries.

With the development of Transformer technology, it has been widely used in the field of cross-modal retrieval. Transformer is a deep learning model based on a self-attention mechanism, capable of capturing long-distance dependencies in input sequences, and has made significant progress in areas such as natural language processing. In cross-modal retrieval tasks, a Transformer is used to model the relationship between text and images. He et al. [22] proposed an end-to-end RSCMR framework based on vision–language fusion (EnVLF) consisting of two unimodal (vision and language) encoders and a multimodal encoder, which can be optimized by multitask training. Al Rahhal et al. [23] proposed a multilanguage framework based on Transformers for solving the problem of limiting queries to English.

Previous research in RSCMR has predominantly focused on enhancing the utilization of multiscale information [24], improving salient features [25], and achieving semantic alignment [9]. This emphasis is evident in the development of multiscale feature fusion modules and salient feature enhancement techniques. While these approaches have yielded improvements in retrieval performance, they exhibit significant limitations when addressing information-sparse images—characterized by limited target information and complex backgrounds—thus presenting challenges for effective feature extraction and semantic understanding. Current methods often struggle to capture critical information in such contexts, resulting in decreased retrieval accuracy.

Moreover, while some studies have sought to boost model performance through attention mechanisms and feature augmentation [9], [26], [27], these efforts primarily target information-rich image features without specifically optimizing for information-sparse scenarios. For example, certain methods utilize multiscale visual self-attention modules to filter redundant image features and employ cross-modal guidance protocols to strengthen multimodal representations. However, these approaches still face challenges when dealing with information-sparse images, as they inadequately address the inherent imbalance between sparse targets and overwhelming background interference.

B. Dynamic Network

The concept of dynamic neural networks has emerged as a prominent research area in the field of deep learning. In contrast to static models that maintain fixed computational graphs and parameters during the inference stage, dynamic networks can adjust their structures or parameters based on varying inputs. This dynamic adaptability offers significant advantages in terms of accuracy, computational efficiency, adaptability, and more. Lin et al. [28] utilized a Markov process to establish a dynamically pruned network. Li et al. [29] addressed the challenge of scale variance in semantic representation, which is called dynamic routing. The proposed framework generates data-dependent routes that adapt to the scale distribution of each image. Qu et al. [30] applied the dynamic network to cross-modal retrieval, enhancing the cross-modal retrieval model's performance by integrating different routing choices.

To the best of our knowledge, the dynamic mechanism has not been previously investigated in the field of RS cross-modal retrieval. Differing from current related studies, our work is inspired by [31] and [32], which introduce a dynamic mechanism to solve the problem of sparse information.

C. Information-Sparse Images

Information-sparse images, characterized by limited key features in low-texture regions, sparse scenes, or low-light conditions, present significant challenges to both traditional and deep learning-based vision systems. Early approaches focused on statistical feature modeling, such as local autocorrelation analysis [33] and sparse representations [34], while frequency-domain methods like wavelet transforms aimed to uncover latent patterns in low-frequency regions. However, these techniques struggled with semantic ambiguity and scalability. Recent advances have leveraged deep learning to address these limitations: dynamic filtering architectures adaptively enhance local feature extraction, while graph-based models optimize long-range dependency modeling by reducing redundant connections. Multimodal frameworks like BLIP-2 [35] further enhance sparse visual data with textual context, and contrastive learning strategies [36] improve feature discriminability through data augmentation. Despite these advancements, challenges persist in balancing computational efficiency with adaptive multiscale fusion, especially for high-resolution applications in RS and autonomous systems. This article addresses the problem of information-sparse image retrieval by using text as an information anchor to analyze the sparsity of images.

D. Information Fusion in RSCMR

The application of information fusion techniques in cross-modal retrieval has a long history. In recent years, some works

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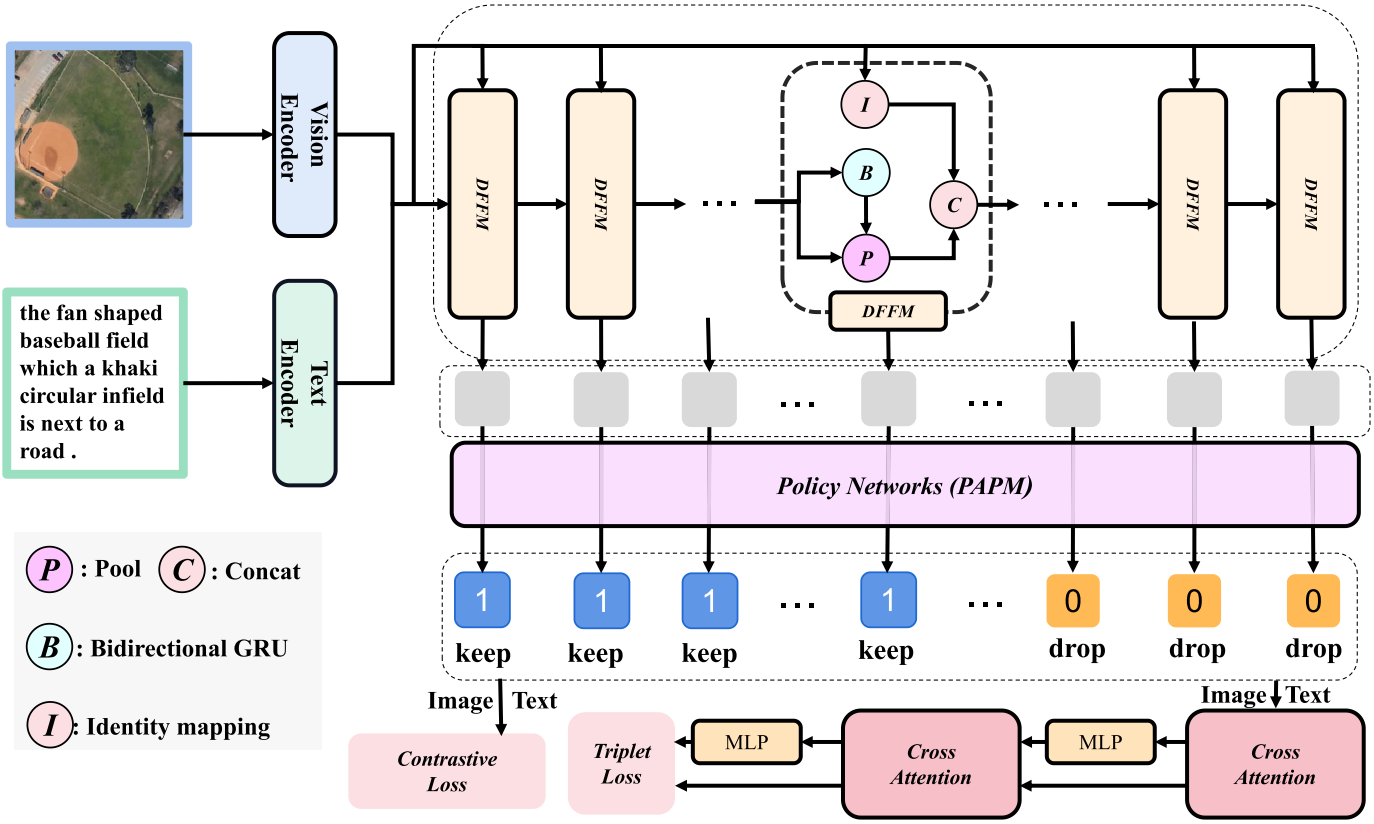


Fig. 2. Schematic illustration of our proposed SIPN. The dynamic network consists of eight layers of DFFN. The policy network, PAPM, determines whether to discard each DFFN layer. The comparison loss is calculated using the image and text features obtained by the dynamic network. Additionally, these features are aligned at multiple levels by an MLAM, which consists of two CAs and two MLPs, and triplet losses are calculated.

have extended the use of information fusion techniques to the field of cross-modal RS retrieval. To address the problem of insufficient utilization of semantic information between different modalities, [37] introduced modal adversarial learning and intramodal semantic discrimination techniques. Li et al. [8] introduced a novel fusion-based contrastive learning model (FBCLM) for RSITR to address the challenges of unimodal feature extraction and correlation exploration of RS image–text pairs. Their model is effective for image–text retrieval on both English and Chinese caption datasets. Lv et al. [38] focused on the issue of complementary semantic information between different modalities and proposed a novel fusion-based correlation learning model (FCLM) for image–text retrieval in RS. Specifically, they designed a cross-modal fusion network to capture the intermodality complementary information and fused features. Similarly, Yao et al. [39] also recognized the challenges posed by RS images, including multiple scales, multiple targets, and rich information. They proposed a method to learn intramodal feature fusion and intermodal feature association to avoid poor retrieval performance caused by information asymmetry. However, they do not solve the problem of cross-modal retrieval of RS images with sparse information.

Building upon insights from prior research, we have developed a DFFM specifically designed to iteratively integrate global and local features, thereby enhancing adaptability to RS images characterized by diverse information densities. Furthermore, in response to the multiscale characteristics inherent in RS images, we have introduced the MLAM. The MLAM harnesses cross-level cross-attention mechanisms to seamlessly integrate features from various levels, facilitating efficient feature extraction across a spectrum of RS image scales.

III. Method

This section introduces the proposed SIPN. Fig. 2 shows the overall architecture.

A. Overall Framework

The architecture of the SIPN is shown in Fig. 2. Suppose we have a set of RS images and text pairs $(\mathcal{V}, \mathcal{T})$, where $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ and $\mathcal{T} = \{t_1, t_2, \ldots, t_N\}$ indicate RS images and texts, respectively, and N denotes the volume of the set. Their relationship labels are $\mathcal{M} = \{m_{ij}\}_{N\times N}$, where $m_{ij} = 1$ means that v_i and t_j contain similar concepts, while $m_{ij} = 0$ implies that they have different concepts. The vision encoder and the text encoder aim to map the RS image and text pairs $(\mathcal{V}, \mathcal{T})$ into d-dimensional feature embeddings. The dynamic network, composed of eight DFFMs, serves as the core of the entire architecture. Its purpose is to construct different training paths for various remote sensing image–text instances. The selection and discarding of specific paths are controlled by the policy network PAPM. To enhance the model's feature selection and

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fusion capabilities, we integrate an MLAM, which consists of CAs and MLPs. The optimization function of the entire network combines contrast loss and triplet loss. Subsequently, we proceed to elaborate on each module extensively.

B. Feature Extraction

This section introduces the feature extractor of the SIPN, which includes a vision encoder and a text encoder.

1) Vision Encoder: The vision encoder efficiently produces multigrained visual concept representations in an image. The encoder is based on Vision Transformer [40]. Initially, an image v whose height, width, and channel are H, W, and C, respectively, is partitioned into n nonoverlapping patches $\mathbf{P} = \{p_1, \ldots, p_n\} \in \mathbb{R}^{(h\times w\times C)\times n}$, where h and w denote the heights and widths of patches, respectively, and $h \times w \times n = H \times W$. Subsequently, we subjected $\mathbf{P}$ to linear embedding. These processed patches are subsequently input into the vision encoder, yielding a set of vectors. The formulation of this step is

$$\mathbf{F} = \mathbf{W}_I \cdot \mathbf{P} = [\mathbf{v}_1, \ldots, \mathbf{v}_n] \tag{1}$$

where $\mathbf{v}_i \in \mathbb{R}^{d\times 1}$ means the ith RS image patches' visual feature. $\mathbf{W}_I \in \mathbb{R}^{d\times(h\times w\times C)}$ means the linear projection matrix. And $\mathbf{F} \in \mathbb{R}^{d\times n}$ means the RS image's local feature. Then, the average of the local features $\mathbf{F}$ is calculated to serve as the initial representation of the entire visual concept, denoted as $\mathbf{v}_0 \in \mathbb{R}^{d\times 1}$. The vision encoder then creates $n + 1$ concept representations $\mathbf{F} \in \mathbb{R}^{d\times(n+1)}$, represented as

$$\mathbf{v}_0 = \mathbb{E}(\mathbf{F}) \tag{2}$$

$$\mathbf{V} = [\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_n] \tag{3}$$

where $\mathbb{E}$ is the expectation.

2) Text Encoder: Similar to the visual encoder, the text encoder creates multigrained visual text. We assume that a text t contains m words, which can be expressed as $\mathbf{W} = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$. We using a dictionary, encode $\mathbf{W}$ into one-hot encoding $\mathbf{O} \in \mathbb{R}^{d_o\times l}$, d_o is the dictionary size, l is the maximum length of the sentence, and then embed $\mathbf{O}$ into text features $\mathbf{G} \in \mathbb{R}^{d\times m}$ through a mapping matrix $\mathbf{W}_T \in \mathbb{R}^{d\times d_o}$. This process can be modeled as

$$\mathbf{G} = \mathbf{W}_T \cdot \mathbf{O} = [\mathbf{t}_1, \ldots, \mathbf{t}_m] \tag{4}$$

where $\mathbf{t}_i \in \mathbb{R}^{d\times 1}$ means the ith word' text feature in the sentence. Due to varying sentence lengths, we set a threshold of p. We remove trailing words for sentences that are too long; for sentences that are too short, we pad the end with zero vectors. In this case, $\mathbf{G}$ is represented as

$$\mathbf{G} = [\mathbf{t}_1, \ldots, \mathbf{t}_l, \ldots, \mathbf{0}, \ldots, \mathbf{0}]_{1\times p}. \tag{5}$$

Then, the average of the local features $\mathbf{G}$ is calculated to serve as the initial representation of the entire text concept, denoted as $\mathbf{t}_0 \in \mathbb{R}^{d\times 1}$. The text encoder then creates $m + 1$ concept representations $\mathbf{G} \in \mathbb{R}^{d\times(m+1)}$, represented as

$$\mathbf{t}_0 = \mathbb{E}(\mathbf{G}) \tag{6}$$

$$\mathbf{T} = [\mathbf{t}_0, \mathbf{t}_1, \ldots, \mathbf{t}_m]. \tag{7}$$

C. Dynamic Network

DFFM is a dynamic network that achieves instance-level adaptation to different RS images by dynamically stacking network layers and adjusting parameter pathways. The dynamic network begins with the initialization of parameters, followed by adjusting network parameters or forward paths through certain dynamic strategies. The specific details are as follows.

1) Dynamic Network Initialization: A comprehensive consideration of all information in the image is necessary to enhance the model's perception of images with sparse information density. This involves aggregating the fragmented information through the encoder to obtain more robust image features. The BiGRU processes feature sequences obtained from the Transformer to consolidate fragmented details. When considering the feature sequence associated with each image patch, this sequence serves as the input to the BiGRU model. The BiGRU model processes these feature sequences, generating an implicit representation at each time step. Due to its bidirectional nature, the BiGRU considers the context of each patch, incorporating information from adjacent patches. A global feature representation is generated as a result, encompassing information from all patches. This process is the initialization of a dynamic network and can be expressed as

$$\mathbf{X}_t = \mathrm{FC}(\mathbf{v}_t) \tag{8}$$

$$\overrightarrow{h_t} = \overrightarrow{\mathrm{GRU}}(\mathbf{X}_t, \overrightarrow{h_{t-1}}) \tag{9}$$

$$\overleftarrow{h_t} = \overleftarrow{\mathrm{GRU}}(\mathbf{X}_t, \overleftarrow{h_{t-1}}) \tag{10}$$

$$\mathbf{Y}_t = \overrightarrow{h_t} \oplus \overleftarrow{h_t} \tag{11}$$

where $\overrightarrow{h_t}$ and $\overleftarrow{h_t}$ represent the output of the tth hidden layer, and $\mathrm{FC}(\cdot)$ represent the fully connected layer

$$\delta = \sigma(\mathrm{BN}(\mathrm{FC}(\mathbf{Y}_L))) \tag{12}$$

where L represents the number of layers of BiGRU, $\sigma(\cdot)$ represent the ReLU function, and $\mathrm{BN}(\cdot)$ represent the batch normalization. The feature sequence $\mathbf{F}$, derived from the image, and the feature sequence $\mathbf{G}$, obtained from the sentence, are processed through the initialized dynamic network $\mathrm{DFFM}_0(\cdot)$. The global representations of the image features, δ_v, and the text features, δ_t, are obtained as follows:

$$\delta_v^0 = \mathrm{DFFM}_{\mathrm{init}}(\mathbf{F}) \tag{13}$$

$$\delta_t^0 = \mathrm{DFFM}_{\mathrm{init}}(\mathbf{G}). \tag{14}$$

The resulting enhanced global feature is concatenated with the original feature sequence and then fused with BiGRU. In cases where image data contains abundant information, the original feature sequence serves as a more suitable representation. However, the enhanced global feature excels as a visual representation for sparsely informative images. Additionally, recognizing the different levels of information in images, the fusion of these two elements involves using weights. The selection of these weights depends on the number of overlapping feature fusion layers, which will be explained in more detail in the following sections. The output can be represented as follows:

$$\mathbf{F}^E = \mathrm{cat}\left(\delta_v^0, \mathbf{F}\right) = \left[\delta_v^0, \mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_n\right] \tag{15}$$

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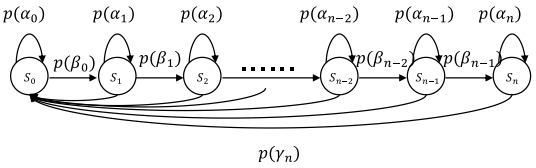


Fig. 3. Dynamic learning with an illustration of model states.

$$\mathbf{G}^E = \text{cat}\left(\delta_t^0, \mathbf{G}\right) = \left[\delta_t^0, \mathbf{t}_0, \mathbf{t}_1, \ldots, \mathbf{t}_m\right] \tag{16}$$

$$\mathbf{F}_0 = \left[\delta_v^0, \mathbf{v}_0\right], \quad \mathbf{G}_0 = \left[\delta_t^0, \mathbf{t}_0\right] \tag{17}$$

where cat(·) means to concatenate two vectors and $\mathbf{F}_0$ and $\mathbf{G}_0$ indicate that no dynamic stacking has occurred.

2) Dynamic Network Transfer: So far, we have derived embeddings for textual and visual content encompassing text and RS images. Then, the alignment of $\mathbf{F}_0$ and $\mathbf{G}_0$ has enabled cross-modal retrieval of RS images. To accommodate images with varying levels of information content, we developed a dynamic stacking feature fusion layer that allows for dynamic feature fusion.

A diagram illustrating the dynamic stacking process is presented in Fig. 3. S_i denotes that the model has stacked i layers of feature fusion layers, which means $S_i = \text{DFFM}_i(\cdot)$, and its transition probability matrix can be represented as follows:

$$P = \begin{pmatrix} p(\alpha_0) & p(\beta_0) & 0 & \cdots & 0 \\ p(\gamma_1) & p(\alpha_1) & p(\beta_1) & \cdots & 0 \\ p(\gamma_2) & 0 & p(\alpha_2) & \cdots & 0 \\ p(\gamma_3) & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ p(\gamma_{n-1}) & 0 & 0 & \cdots & p(\beta_{n-1}) \\ p(\gamma_n) & 0 & 0 & \cdots & p(\alpha_n) \end{pmatrix} \tag{18}$$

where α represents the identity transfer, signifying the feature fusion layer without any superposition; β signifies the addition of a layer of feature fusion layer; and γ denotes the removal of all feature fusion layers, effectively returning to the initial state. $p(\alpha_i)$, $p(\beta_i)$, and $p(\gamma_i)$ represent the probabilities of α, β, and γ transitions occurring in the ith step, respectively. The specific process can be represented using the following modeling:

$$D_V = \{\text{DFFM}_j(\mathbf{F}_j) | j = 0, 1, \ldots, n\} \tag{19}$$

$$D_T = \{\text{DFFM}_j(\mathbf{G}_j) | j = 0, 1, \ldots, n\} \tag{20}$$

$$(D_V, D_T) \times P \rightarrow (D_V, D_T) \tag{21}$$

$$\text{DFFM}_j(\mathbf{F}_j) = \begin{cases} \displaystyle\sum_{i=1}^{n} \frac{p(\gamma_i)\text{DFFM}_i(\mathbf{F}_i)}{1 - p(\alpha_0)}, & j = 0 \\ \dfrac{p(\beta_{j-1})\text{DFFM}_j(\mathbf{F}_{j-1})}{1 - p(\alpha_j)}, & j \neq 0 \end{cases} \tag{22}$$

$$\text{DFFM}_j(\mathbf{G}_j) = \begin{cases} \displaystyle\sum_{i=1}^{n} \frac{p(\gamma_i)\text{DFFM}_i(\mathbf{G}_i)}{1 - p(\alpha_0)}, & j = 0 \\ \dfrac{p(\beta_{j-1})\text{DFFM}_j(\mathbf{G}_{j-1})}{1 - p(\alpha_j)}, & j \neq 0 \end{cases} \tag{23}$$

where n is the number of network layers, $\mathbf{F}_i$ and $\mathbf{F}_j$ represent the features obtained after stacking i and j layers of feature fusion layers from the image, respectively, and $\mathbf{G}_i$ and $\mathbf{G}_j$ represent the corresponding features from the sentence. $\text{DFFM}_i(\cdot)$ and $\text{DFFM}_j(\cdot)$ represent the ith and jth layer of dynamic networks, respectively. D_V and D_T are finite state sets of visual and textual information. The final output is expressed as follows:

$$\mathbf{F}_j = \left[\delta_v^j, \mathbf{v}_0^j\right], \quad \mathbf{G}_j = \left[\delta_t^j, \mathbf{t}_0^j\right] \tag{24}$$

where δ_v^j and δ_t^j represent the global dynamic features of the jth layer after DFFM processing; $\mathbf{v}_0^j$ and $\mathbf{t}_0^j$ represent the local dynamic features of the jth layer after DFFM processing.

D. Policy Network

PAPM is a pretrained *policy network* and does not participate in subsequent training phases. It is a meticulously trained binary classifier on the MSCOCO [41] dataset, intended to effectively differentiate between image and text features. The model employs the binary cross-entropy loss function, adept at quantifying the dissimilarities between the two modalities. The model consists of an MLP and a sign function. The PAPM uses a cross-modal pretraining model to perform the binary classification task of image and text features. Through an extensive pretraining phase, the model becomes adept at capturing intricate nuances and unique characteristics that distinguish image and text features. Consequently, the PAPM model demonstrates exceptional proficiency in accurately discerning and classifying distinct modalities. This process can be modeled as follows:

$$\mathcal{V}_{\text{coco}} = f(I), \mathcal{T}_{\text{coco}} = g(T) \tag{25}$$

$$\hat{y} = \text{MLP}\left(\left\{\mathbf{v}_0^{\text{coco}}, \mathbf{v}_1^{\text{coco}}, \ldots, \mathbf{v}_n^{\text{coco}}; \mathbf{t}_0^{\text{coco}}, \mathbf{t}_1^{\text{coco}}, \ldots, \mathbf{t}_n^{\text{coco}}\right\}\right) \tag{26}$$

$$\mathcal{L}_{\text{mis}} = -\left(y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})\right) \tag{27}$$

where I and T are images and text in MSCOCO, respectively. $f(\cdot)$ is an image feature extractor initialized with Swin Transformer, and $g(\cdot)$ is a text feature extractor initialized with BERT. $\mathbf{v}_i^{\text{coco}}$ is the ith image feature in $\mathcal{V}_{\text{coco}}$, $\mathbf{t}_i^{\text{coco}}$ is the ith image feature in $\mathcal{T}_{\text{coco}}$, y is the correct label, y is 1 when the input feature is an image feature and 0 when the input feature is a text feature, and $\hat{y}$ is the model prediction label.

Given the inherent characteristics of the PAPM, we can establish a threshold value of 0.5 to enforce the mapping of image features to the image space and text features to the text space. Applying this threshold ensures that the extracted representations align with their respective modalities. This constraint guarantees that image features are effectively projected within the image space while text features maintain their mapping within the text space. In other words, $\text{sign}(\text{MLP}(\mathcal{V}_{\text{coco}})) = 1$ and $\text{sign}(\text{MLP}(\mathcal{T}_{\text{coco}})) = 0$, where the sign function can be represented as

$$\text{sign}(x) = \begin{cases} 0, & x < 0.5 \\ 1, & x \geq 0.5 \end{cases} \tag{28}$$

where 0 indicates drop and 1 indicates keep. Finally, we can model the process of using a policy network to make judgments as

$$\eta = \text{sign}(\text{PAPM}(x)) \tag{29}$$

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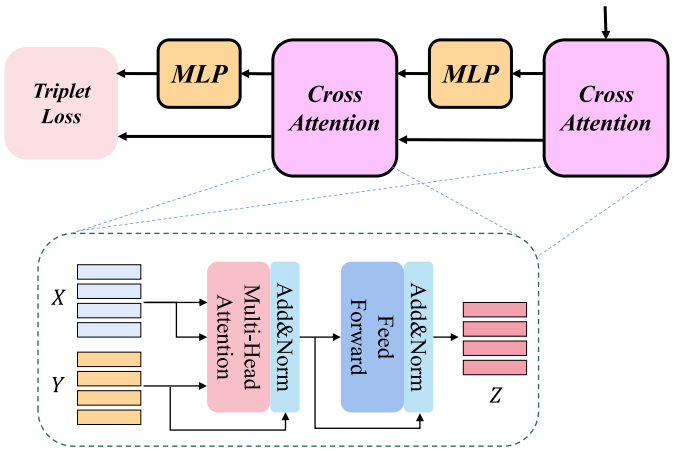


Fig. 4. Illustration of the MLAM. The figure showcases the processing flow of the image component, with the text component following an identical processing procedure.

where PAPM(·) is the trained policy network, x is the features, η is the result, 0 indicates the drop, and 1 indicates the keep. In the end, the features obtained through the DFFM can be represented as

$$\bar{\mathbf{v}}^j = \mathbb{E}\left(\left[\mathbf{v}^0, \ldots, \mathbf{v}^j\right]\right), \quad \bar{\mathbf{t}}^j = \mathbb{E}\left(\left[\mathbf{t}^0, \ldots, \mathbf{t}^j\right]\right) \tag{30}$$

$$\mathbf{v}_l = P^v_{j,0} \cdot \mathbf{v}^0 + \left(1 - P^v_{j,0}\right) \cdot \operatorname{sign}\left(\operatorname{PAPM}\left(\bar{\mathbf{v}}^j\right)\right) \cdot \bar{\mathbf{v}}^j \tag{31}$$

$$\mathbf{t}_l = P^t_{j,0} \cdot \mathbf{t}^0 + \left(1 - P^t_{j,0}\right) \cdot \operatorname{sign}\left(\operatorname{PAPM}\left(\bar{\mathbf{t}}^j\right)\right) \cdot \bar{\mathbf{t}}^j \tag{32}$$

where P^v represents the probability transition matrix for images and P^t represents the probability transition matrix for text.

E. Multilevel Attention Model

This section will explore the MLAM. This module comprises two cross-attention layers and two multilayer perceptron (MLP) layers. Its main objective is to utilize various attention mechanisms to acquire semantic information at different levels. This is particularly important for RS images, which often show significant differences in semantic density. To address this challenge, the MLAM employs cross-attention to establish connections between different levels of semantic information. This enables the model to effectively capture meaningful semantic content, even when dealing with images with limited semantic information. The specific process is visualized in Fig. 4. The module's operation can be simplified as follows: it conducts attention fusion between the high-level features generated by the DFFM and the original low-level features. By integrating information from different levels through cross-attention, the MLAM enhances the model's capacity to capture diverse and rich semantic information, improving its performance in image–text retrieval tasks. The specific process can be described as follows:

$$\mathbf{v}_l^* = \operatorname{MLP}(\operatorname{CA}(\mathbf{v}_l, \delta_v)) \tag{33}$$

$$\mathbf{v}_g^* = \operatorname{MLP}\left(\operatorname{CA}\left(\phi\left(\mathbf{v}_l\right), \mathbf{v}_l^*\right)\right) \tag{34}$$

where MLP() means the operations of MLPs, $\phi()$ represents the operation of extracting the CLS token from the feature sequence and replicating it to fill the entire sequence up to the maximum length, and CA() stands for cross-attention. The specific process can be expressed as

$$\begin{aligned} Z &= \operatorname{CA}(X, Y) \\ &= \operatorname{softmax}\left(\frac{\mathbf{W}^Q(Y) \cdot \mathbf{W}^K(X)^T}{\sqrt{d_K}}\right) \cdot \mathbf{W}^V(X) \end{aligned} \tag{35}$$

where $\mathbf{W}^Q()$, $\mathbf{W}^K()$, and $\mathbf{W}^V()$ denote three projection matrices that map the normalized input data Y into queries and data X into keys and values, d_k indicates a scaled factor, and softmax() is the softmax function.

F. Objective Function

This article needs to calculate the loss of three sets of features, namely $\mathcal{L}_g$, $\mathcal{L}_l^*$, and $\mathcal{L}_g^*$, which represent three levels of features for images and text, respectively. Among them, $\mathcal{L}_g$ is calculated by contrast loss, and $\mathcal{L}_l^*$ and $\mathcal{L}_g^*$ are calculated by triplet loss.

The objective function $\mathcal{L}_g$ involves computing the contrast loss for text and visual features obtained through the DFFM. This computation aims to enhance the model's ability to perceive images characterized by sparse information. The objective function $\mathcal{L}_l^* + \mathcal{L}_g^*$ pertains to calculating the triplet loss for multilevel text features and visual features acquired via the MLAM. This computation aims to enhance the model's ability to match multiscale images with corresponding text descriptions.

We can more effectively model cross-modal relationships by optimizing the similarity and dissimilarity of sample pairs using contrastive loss. The contrast loss can be expressed as

$$\mathbb{F}(\mathcal{V}, \mathcal{T}) = -\log \frac{\exp(S(v, t^+)/\tau)}{\sum_{v \in \mathcal{V}, t \in \mathcal{T}} \exp(S(v, t)/\tau)} \tag{36}$$

$$\mathbb{F}(\mathcal{T}, \mathcal{V}) = -\log \frac{\exp(S(t, v^+)/\tau)}{\sum_{v \in \mathcal{V}, t \in \mathcal{T}} \exp(S(t, v)/\tau)} \tag{37}$$

where τ is the temperature coefficient. By optimizing the relative distances between samples using a triplet loss, we aim to enhance the feature embeddings of images and text more precisely within the shared space. The triple loss can be expressed as

$$\mathbb{G}(\mathcal{V}, \mathcal{T}) = \sum_{v \in \mathcal{V}, t^+, t^- \in \mathcal{T}} [\text{margin} - S(v, t^+) + S(v, t^-)]_+ \tag{38}$$

$$\mathbb{G}(\mathcal{T}, \mathcal{V}) = \sum_{v \in \mathcal{V}, t^+, t^- \in \mathcal{T}} [\text{margin} - S(t, v^+) + S(t, v^-)]_+ \tag{39}$$

where (v, t^+) and (t, v^+) are paired sample pairs, (v, t^-) and (t, v^-) are not paired sample pair, and "margin" is the threshold for triplet loss.

$\mathcal{L}_g$, $\mathcal{L}_l^*$, and $\mathcal{L}_g^*$ can be expressed as

$$\mathcal{L}_g = \frac{1}{2} \times \left(\mathbb{F}\left(\mathcal{V}_g, \mathcal{T}_g\right) + \mathbb{F}\left(\mathcal{T}_g, \mathcal{V}_g\right)\right) \tag{40}$$

$$\mathcal{L}_l^* = \frac{1}{2} \times \left(\mathbb{G}\left(\mathcal{V}_l^*, \mathcal{T}_l^*\right) + \mathbb{G}\left(\mathcal{T}_l^*, \mathcal{V}_l^*\right)\right) \tag{41}$$

$$\mathcal{L}_g^* = \frac{1}{2} \times \left(\mathbb{G}\left(\mathcal{V}_g^*, \mathcal{T}_g^*\right) + \mathbb{G}\left(\mathcal{T}_g^*, \mathcal{V}_g^*\right)\right) \tag{42}$$

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where $\mathcal{V}_g$, $\mathcal{V}_l^*$, and $\mathcal{V}_g^*$ represent the sets of v_g, v_l^*, and v_g^*, respectively. $\mathcal{T}_g$, $\mathcal{T}_l^*$, and $\mathcal{T}_g^*$ represent the sets of t_g, t_l^*, and t_g^*, respectively. The final loss can be expressed as

$$\mathcal{L} = \mathcal{L}_g + \alpha \times \left(\mathcal{L}_l^* + \mathcal{L}_g^*\right). \tag{43}$$

In the above formula, α is a hyperparameter.

IV. Experimental Results and Analysis

In this section, we first introduce the dataset, evaluation indicators, and experiment details. Hereafter, we conduct comparisons and analyses on several RS text–image datasets to verify the effectiveness of the devised SIPN. Furthermore, plenty of ablation experiments have been carried out to explore the reasons for the excellent performance of the SIPN systematically. Then, the PAPM and the hyperparameters of the DFFM layers are studied. Finally, we conduct hyperparameter analysis, retrieval visualization, and attention heatmap visualization.

A. Datasets and Evaluation Metrics

1) Datasets: In our study, we evaluate the performance of our proposed method on two benchmark RS datasets: RSICD [21] and RSITMD [10]. We split the datasets into 80% for training, 10% for validation, and the remaining 10% for testing.

a) RSICD: RSICD [21] is a comprehensive dataset designed explicitly for generating captions for RS images. It consists of 10 921 images and corresponding captions, making it a valuable resource for advancing research in RS and natural language processing. The dataset encompasses a wide range of geographic environments, including urban areas, rural landscapes, agricultural fields, mountains, and bodies of water. These images are sourced from Google Earth and Google Street View. Each image is accompanied by five manually annotated English captions, providing detailed image content descriptions. The images in the dataset have been standardized to a resolution of 224 × 224 pixels, ensuring consistency across the dataset.

b) RSITMD: RSITMD [10] is a comprehensive and challenging dataset that facilitates RS image retrieval based on keywords and sentences, either separately or jointly. It combines images from the RSICD dataset and Google Earth, resulting in a diverse collection. The dataset consists of 4743 images, for which 23 715 captions are provided. This dataset is a valuable resource for studying fine-grained RS image retrieval. Researchers can explore various retrieval techniques that utilize both textual and visual information to improve the accuracy and effectiveness of retrieval systems in the RS domain.

2) Evaluation Metric: We conduct two kinds of image–text matching tasks: 1) sentence retrieval, that is, retrieving ground-truth sentences related to the query image (I2T) and 2) image retrieval, that is, retrieving ground-truth images related to the query text (T2I). Based on [48], $R@k$ and mR [54] are used as evaluation criteria to assess the model performance. $R@k$ aims to calculate the proportion of ground truth in the top k samples recalled. In the experiment, k is taken as 1, 5, and 10 to evaluate the results more reasonably. The mR is then leveraged to represent the average of multiple $R@k$, which reflects the model's overall performance more intuitively. The mR is defined as follows:

$$\text{mR} = \frac{1}{6} \times (\underbrace{R@1 + R@5 + R@10}_{\text{Image-to-Text}} + \underbrace{R@1 + R@5 + R@10}_{\text{Text-to-Image}}). \tag{44}$$

B. Implementation Details

All experiments in this study were conducted using four NVIDIA RTX A6000 GPUs. For consistency across different datasets with varying image sizes, we uniformly resized them to 224 × 224 pixels and input them into the network. Data augmentation techniques, including rotation and flip, were applied to enhance model robustness. The word vector's representation dimension was 768, while the embedding space for images and text was 512. The parameter τ was initialized to 0.07 and learned during contrast loss computation, and a margin of 0.5 was enforced for triplet loss computation. The network was trained for 20 epochs using the AdamW optimizer with both contrast loss and triplet loss, and a batch size of 128 per GPU was used. To ensure the robustness of the experimental results, each set of experiments was conducted five times, and the average result was recorded.

C. Comparisons With SoTA Methods

We compare the proposed model with the multiple progressive cross-modal retrieval models on two RS text–image datasets. The SIPN demonstrates excellent performance on both datasets: RSICD and RSITMD. Through comprehensive evaluations, we have consistently achieved impressive results, outperforming existing methods in accuracy and efficiency. The robustness and versatility of our model make it well-suited for various cross-modal image–text retrieval tasks. Experimental findings confirm the effectiveness and reliability of our approach, establishing it as a state-of-the-art solution in RS data analysis.

1) Experimental Results on the RSICD Dataset: The results on the RSITMD dataset are shown in Table I. Our method outperforms all state-of-the-art methods in overall performance. It can be seen that MT-ENG performs the best among all other methods, but our method can still outperform MT-ENG. Specifically, compared with SWAN, our model can obtain an increase of 2.46% on the mR metric, which shows the superiority of our model.

In sentence retrieval, our model has achieved the best results on all indicators. Our proposed method yields 14.00%(R@1), 28.76%(R@5), and 40.04%(R@10), and it outperforms MT-ENG by over 3.30%(R@1), 1.28%(R@5), and 3.39%(R@10).

Our model has achieved the best results on all indicators in image retrieval. Our proposed method yields 9.86%(R@1), 31.25%(R@5), and 48.31%(R@10), and it outperforms MT-ENG by over 0.72%(R@1), 2.29%(R@5), and 3.72%(R@10).

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TABLE I
COMPARISONS OF RETRIEVAL PERFORMANCE ON THE RSICD TEST SET

Method	Backbone vision encoder / text encoder	Sentence Retrieval R@1	R@5	R@10	Image Retrieval R@1	R@5	R@10	mR
VSE++ [42]	ResNet-50 / GRU	3.38	9.51	17.46	2.82	11.32	18.10	10.43
SCAN [43]	RoI Transformer / GRU	5.85	12.89	19.84	3.71	16.40	26.73	14.23
CAMP [44]	RoI Transformer / GRU	5.12	12.89	21.12	4.15	15.23	27.81	14.39
MSITA [45]	Vision Transformer / GRU	8.67	22.71	33.91	6.13	21.98	5.39	21.47
MTFN [46]	CNN + Faster-RCNN / GRU	5.02	12.52	19.74	4.90	17.17	29.49	14.81
LW-MCR [47]	CNN Blocks / CNN Blocks	3.29	12.52	19.93	4.66	17.51	30.02	14.66
AMFMN [21]	ResNet-18 / GRU	5.39	15.08	23.40	4.90	18.28	31.44	16.42
GaLR [48]	ResNet-18 + ppyolo / GRU	6.59	19.85	31.04	4.69	19.48	32.13	18.96
SWAN [49]	ResNet-50 / GRU	7.41	20.13	30.86	5.56	22.26	37.41	20.61
VGSGN [50]	CNN / GRU	8.33	21.87	32.57	6.53	23.13	36.85	21.55
DOVE [51]	ResNet-50 / GRU	8.66	22.35	34.95	6.04	23.95	40.35	22.72
EnVLF [22]	Vision Transformer / BERT	7.78	20.52	31.56	6.09	22.33	6.65	20.82
SMLGN [51]	Vision Transformer / BERT	8.87	25.53	37.24	7.85	27.14	42.58	24.87
PIR [52]	Swin Transformer + ResNet-50 / Bert	9.88	27.26	39.16	6.87	24.56	38.92	24.46
MT-ENG [23]	Dual Transformer	10.70	29.64	41.53	9.14	28.96	44.59	27.42
SIPN	Swin Transformer / Bert	**14.00**	**30.92**	**44.92**	**9.86**	**31.25**	**48.31**	**29.88**

TABLE II
COMPARISONS OF RETRIEVAL PERFORMANCE ON THE RSITMD TEST SET

Method	Backbone vision encoder / text encoder	Sentence Retrieval R@1	R@5	R@10	Image Retrieval R@1	R@5	R@10	mR
VSE++ [42]	ResNet-50 / GRU	10.38	27.65	39.60	7.79	24.87	38.67	24.83
SCAN [43]	RoI Transformer / GRU	10.18	28.53	38.49	10.10	28.98	43.53	26.64
CAMP [44]	RoI Transformer / GRU	11.73	26.99	38.05	8.27	27.79	44.34	26.20
MSITA [45]	Vision Transformer / GRU	15.22	34.20	47.65	12.15	39.92	57.72	34.48
MTFN [46]	CNN + Faster-RCNN / GRU	10.40	27.65	36.28	9.96	31.37	45.84	26.92
LW-MCR [47]	CNN Blocks / CNN Blocks	10.18	28.98	39.82	7.79	30.18	49.78	27.79
AMFMN [21]	ResNet-18 / GRU	10.63	24.78	41.81	11.51	34.69	54.87	29.72
GaLR [48]	ResNet-18 + ppyolo / GRU	13.05	30.09	42.70	10.47	36.34	53.35	31.00
SWAN [49]	ResNet-50 / GRU	13.35	32.15	46.90	11.24	40.40	60.60	34.11
VGSGN [50]	CNN / GRU	14.16	34.96	50.66	13.23	42.57	63.41	36.50
DOVE [51]	ResNet-50 / GRU	16.81	36.80	49.93	12.20	44.13	66.50	37.73
EnVLF [22]	Vision Transformer / BERT	13.42	33.11	47.12	11.15	38.63	57.58	33.50
SMLGN [53]	Vision Transformer / BERT	17.26	39.38	51.55	13.19	43.94	60.40	37.62
PIR [52]	Swin Transformer + ResNet-50 / Bert	18.14	41.15	52.88	12.17	41.68	63.41	38.24
MT-ENG [23]	Dual Transformer	19.69	40.26	54.42	**17.61**	49.73	66.59	41.38
SIPN	Swin Transformer / Bert	**23.01**	**46.02**	**57.96**	17.04	**49.87**	**66.95**	**43.47**

2) Experimental Results on the RSITMD Dataset: The results on the RSITMD dataset are shown in Table II. Our method outperforms all state-of-the-art methods in overall performance. It can be seen that MT-ENG performs the best among all other methods, but our method can still outperform MT-ENG, except for image retrieval R@1. Specifically, compared with MT-ENG, our model can obtain an increase of 2.09% on the mR metric, which shows the superiority of our model.

In sentence retrieval, our model has achieved the best results on all indicators. Our proposed method yields 23.01%(R@1), 46.02%(R@5), and 57.96%(R@10), and it outperforms MT-ENG by over 3.32%(R@1), 5.76%(R@5), and 3.54%(R@10).

In image retrieval, our model has achieved the best results on all indicators except for R@1. Our proposed method yields 17.04%(R@1), 49.87%(R@5), and 66.95%(R@10), and it outperforms MT-ENG by over −0.57%(R@1), 0.14%(R@5), and 0.36%(R@10).

D. Effective Study on the DFFM and MLAM

We conduct detailed experiments in this section to analyze the proposed SIPN systematically. As shown in

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TABLE III
ABLATION STUDY ON THE RSICD DATASETS

Method	Text DFFM	Vision DFFM	MLAM	Sentence Retrieval			Image Retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
NET-1	✗	✗	✗	9.61	26.62	38.15	6.84	22.71	36.21	23.36
NET-2	✓	✗	✗	9.70	26.62	39.71	7.92	26.59	41.96	25.42
NET-3	✓	✓	✗	10.98	29.55	43.00	8.34	28.18	43.51	27.26
NET-4	✓	✓	✓	**14.00**	**30.92**	**44.92**	**9.86**	**31.25**	**48.31**	**29.88**

TABLE IV
ABLATION STUDY ON THE RSITMD DATASETS

Method	Text DFFM	Vision DFFM	MLAM	Sentence Retrieval			Image Retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
NET-1	✗	✗	✗	17.04	38.50	53.32	12.74	42.74	65.00	32.22
NET-2	✓	✗	✗	21.24	42.48	54.20	16.77	44.07	60.35	39.85
NET-3	✓	✓	✗	21.02	44.03	57.74	16.02	45.71	64.07	41.43
NET-4	✓	✓	✓	**23.01**	**46.02**	**57.96**	**17.04**	**49.87**	**66.95**	**43.47**

Tables III and IV, we constructed four networks to explore the contributions of different components to the SIPN.

1) *NET-1:* vision encoder + text encoder.
2) *NET-2:* vision encoder + text encoder + text DFFM.
3) *NET-3:* vision encoder + text encoder + text DFFM + vision DFFM.
4) *NET-4:* vision encoder + text encoder + text DFFM + vision DFFM + MLAM.

Here, NET-1 contains only a vision encoder and a text encoder, and the loss function is just a contrast loss. NET-2 and NET-3 are added to the DFFM. NET-2 only adds the DFFM to the text processing, while NET-3 adds the DFFM to both the text and image processing. NET-4 is an SIPN. Here, the hyperparameters used in the model are consistent, including the settings of the PAPM and DFFM layers.

We can easily see that the values of all assessment criteria have increased from NET-1 to NET-4. For example, compared with NET-1, the significant enhancements in mR achieved by NET-4 are 6.52% (RSICD) and 11.25% (RSITMD). Taking the "sentence retrieval" results of RSITMD as another example, the improvements in R@1 obtained by NET-4 are 5.97% (NET-1), 1.77% (NET-2), 1.99% (NET-3), respectively. The above discussion confirms the contributions of the DFFM and MLAM to the SIPN. Furthermore, we can summarize three points from the results.

1) Despite the notable performance disparity between NET-3 and NET-2, the DFFM demonstrates its capacity to enhance the model's proficiency in processing sparse RS images. This finding underscores the pressing necessity to investigate and address the challenge of sparse information in RS images.
2) The difference between NET-1 and NET-2 is the inclusion of the DFFM for feature postprocessing in the textual segment. The significant difference in performance between the RSICD and RSITMD datasets indicates that the textual data in these datasets may lack adequate information or suffer from insufficient feature extraction. This aspect deserves further investigation.

TABLE V
ABLATION STUDY ON THE LAYER-WISE CONFIGURATION OF THE MLAM

Layers	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
1	22.33	44.51	56.71	16.87	47.19	**67.01**	42.44
2	**23.01**	**46.02**	**57.96**	**17.04**	**49.87**	66.95	**43.47**
3	19.89	42.43	55.89	15.9	48.12	66.23	41.41
4	19.76	42.30	55.71	15.78	48.00	66.09	41.27

TABLE VI
HYPERPARAMETER SENSITIVITY EXPERIMENT IN RSITMD

α	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
0.001	20.55	42.60	55.48	15.03	46.70	63.81	41.18
0.01	21.80	44.15	56.22	15.50	48.70	65.03	41.97
0.1	**23.05**	46.00	57.93	**17.08**	49.85	**66.98**	43.48
1	23.03	**46.06**	**57.95**	17.05	**49.88**	66.96	**43.49**
10	22.96	46.03	57.97	17.03	49.86	66.94	43.47
100	19.85	43.22	54.01	13.08	46.15	62.98	40.28

3) The incorporation of the MLAM into NET-3 results in the formation of NET-4. The performance of NET-4 surpasses that of NET-3, signifying the substantive contribution of the MLAM in addressing sparse information issues in RS images. Furthermore, multilevel triplet learning enhances the model's retrieval performance.

E. MLAM Layer Experiment

We analyzed the effect of different MLAM layers on the model performance. The results are presented in Table V. Our experimental investigation, which varies the layer configurations of the MLAM, reveals that optimal performance is achieved with a two-layer stacked architecture. Empirical results demonstrate that deeper stacking beyond two layers induces overfitting during the training phase, thereby

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TABLE VII
EXPERIMENTS ON THE PAPM AND DFFM LAYERS ON THE RSICD TEST SET

PAPM	DFFM Layers	Sentence Retrieval			Image Retrieval			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
✗	1	10.61	28.18	38.98	8.25	27.28	42.89	26.03
✗	2	10.98	29.46	41.26	8.36	28.51	44.87	27.24
✗	3	11.16	28.91	41.90	8.47	28.73	44.32	27.25
✗	4	12.81	27.81	41.17	7.67	27.59	44.70	26.96
✗	5	11.53	29.37	40.71	7.70	26.57	43.59	26.58
✗	6	10.70	26.90	41.63	8.78	28.58	44.46	26.84
✗	7	12.17	28.73	40.99	8.25	27.63	44.56	27.05
✗	8	13.27	28.18	39.07	8.73	27.28	42.85	26.56
✓	/	**14.00**	**30.92**	**44.92**	**9.86**	**31.25**	**48.31**	**29.88**

TABLE VIII
EXPERIMENTS ON THE PAPM AND DFFM LAYERS ON THE RSITMD TEST SET

PAPM	DFFM Layers	Sentence Retrieval			Image Retrieval			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
✗	1	21.90	41.59	55.97	15.13	45.35	64.60	40.76
✗	2	20.80	43.14	55.09	14.91	43.89	63.85	40.28
✗	3	21.90	41.59	55.97	15.13	45.35	64.60	40.76
✗	4	16.81	43.81	56.42	14.03	43.76	65.40	40.04
✗	5	16.81	41.15	55.31	14.91	45.13	66.28	39.93
✗	6	22.12	42.26	53.76	15.88	45.22	63.01	40.38
✗	7	19.91	44.69	58.41	16.24	45.40	65.35	41.67
✗	8	18.81	42.92	53.76	15.97	43.94	64.87	40.04
✓	/	**23.01**	**46.02**	**57.96**	**17.04**	**49.87**	**66.95**	**43.47**

TABLE IX
PERFORMANCE VERIFICATION OF DYNAMIC NETWORKS BUILT USING DIFFERENT MODELS ON RSICD AND RSITMD

Method	**RSICD**							**RSITMD**						
	Sentence Retrieval			**Image Retrieval**			**mR**	**Sentence Retrieval**			**Image Retrieval**			**mR**
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
MLP_0	8.78	24.89	39.98	7.98	26.77	43.17	25.26	16.69	39.82	56.86	14.78	44.60	64.07	39.45
MLP_1	6.50	25.89	40.35	7.52	25.76	42.93	24.82	15.49	39.82	53.98	12.83	45.58	65.58	38.88
BiLSTM	5.95	26.44	39.62	8.09	26.94	42.47	24.92	17.04	38.05	50.88	15.31	44.51	61.28	37.85
BiGRU	7.32	23.97	37.88	7.69	27.30	42.76	24.49	13.50	36.73	49.56	12.92	43.50	63.58	36.63
$DFFM_0$	8.14	26.99	39.25	8.31	28.25	45.69	26.11	18.58	44.69	55.53	14.25	45.27	64.42	40.46

establishing the two-layer configuration as the most effective implementation.

F. Hyperparameter Sensitivity Analysis

We analyzed the impact of different hyperparameter values of α on model performance. The results are shown in Table VI. Experimental findings indicate that the model performs relatively stable when α is in the range of 0.1–10, with minimal fluctuations in evaluation metrics. This suggests that the model exhibits a certain degree of robustness in this range. However, when α is less than 0.1, the regularization may be too weak, leading to overfitting and a significant drop in performance. Conversely, when α exceeds 10, excessive regularization may hinder the model's learning capacity, also resulting in degraded performance. Therefore, selecting an appropriate value of α is crucial for achieving optimal model performance.

G. Experiments on the PAPM and DFFM Layers on Different Datasets

The impact of the PAPM and DFFM iteration layers on model performance is further investigated. The evaluation of the hyperparameters associated with the PAPM and DFFM layers is presented in Tables VII and VIII. These experiments were conducted using the RSICD and RSITMD datasets, respectively. The results indicate that optimal performance on RSICD was achieved with three layers, whereas seven layers yielded superior performance on RSITMD when the PAPM was not employed to terminate the DFFM cycle. This phenomenon may be attributed to dataset characteristics; specifically, RSICD exhibits a higher level of noise compared to RSITMD. Consequently, an excessive number of DFFMs can result in noise accumulation, adversely affecting model performance. Nevertheless, the DFFM based on policy networks demonstrated optimal results across both datasets, indicating that network dynamic construction is adaptable to varying datasets. Incorporating the PAPM enhanced performance across both datasets, thereby reinforcing its efficacy.

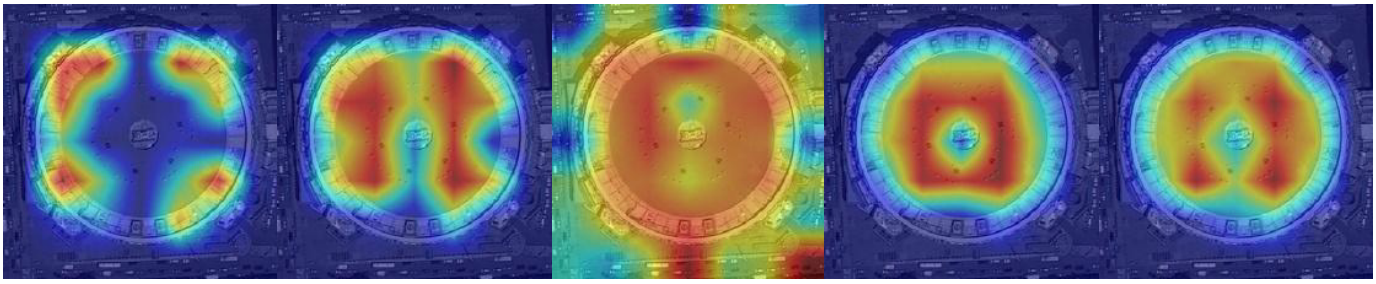
H. Performance Verification of Dynamic Networks Built Using Different Models on RSICD and RSITMD

As shown in Table IX, construct dynamic networks using different models and perform performance verification

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TABLE X
PRETRAIN THE PAPM ON DIFFERENT DATASETS AND EVALUATE ITS RETRIEVAL PERFORMANCE ON RSICD AND RSITMD

Pretrain Dataset	RSICD							RSITMD						
	Sentence Retrieval			Image Retrieval			mR	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
RSICD	8.69	29.46	43.09	7.54	27.23	43.60	26.60	19.69	40.71	55.31	16.28	48.67	69.30	41.66
RSITMD	11.62	27.90	40.62	8.76	28.67	46.16	27.29	20.58	43.36	57.52	16.77	50.09	67.65	42.74
MSCOCO	**14.00**	**30.92**	**44.92**	**9.86**	**31.25**	**48.31**	**29.88**	**23.01**	**46.02**	**57.96**	**17.04**	**49.87**	**66.95**	**43.47**



(a) (b) (c) (d) (e)

Fig. 5. Visual experiment results of the DFFM are presented, where (a)–(d) correspond to DFFM dynamic stacking for 1–4 layers, respectively, and (e) represents the outcomes after PAPM discontinues stacking.

on RSICD and RSITMD. This includes the construction of single-layer MLPs, multilayer MLPs, BiGRU, and BiLSTM. Experimental results indicate that the dynamic network constructed using BiGRU demonstrates the best retrieval performance.

I. Pretrain the Policy Network Using Different Training Sets

Pretrain the PAPM on different datasets and evaluate its retrieval performance on RSICD and RSITMD, as shown in Table X. The pretrained datasets include MSCOCO, RSICD, and RSITMD, with the best retrieval performance achieved after pretraining on MSCOCO. This superior performance is attributed to MSCOCO's more significant number of image–text pairs and more diverse categories of text descriptions.

J. Visual Analysis of the DFFM

Fig. 5 displays the attention heat map for each layer during the dynamic stacking process of feature fusion layers. The first four depict dynamic stacking occurring once to four times, while the final one presents the attention heat map when the PAPM discontinues stacking, detecting overfitting or noise redundancy. By comparing different levels of the DFFM, we observe the model's attention gradually focusing on critical areas, ultimately ceasing the fusion process of the DFFM under the constraints of the PAPM. Fig. 5 comprehensively demonstrates that the DFFM effectively directs the model's attention to crucial information within the remote sensing image. Furthermore, it underscores the guiding influence of the PAPM on the DFFM.

K. Visual Analysis of Recall Results

In this section, we visualize typical experimental results to analyze the performance differences among various retrieval models intuitively. We have selected the baseline models to compare with our model. The visualized results are shown in Fig. 6. It can be intuitively seen that our model performs better in image understanding with highly sparse semantics.

In the task of using image retrieval for text, the query image is a high aerial view of a specific scene, including a grassland, a river, and scattered buildings. By observing the returned results of two models, it can be noticed that the SIPN model correctly returns the top three results, while the last two results, although incorrect, still somewhat match the image's content. On the other hand, the baseline model only recognizes the grassland that occupies most of the image area while ignoring the river and buildings. Intuitively, it can be understood that the SIPN model is better able to capture details in a larger-scale photograph.

In the task of using text retrieval for images, the query text is "Two dark green cracks cross the yellow desert." As humans, we pay attention to the desert and dark green cracks. Both models return correct results in the top five best matches, but it is evident that the SIPN model instills more confidence in learning "green cracks." In the case of images with a desert background, there is often a problem of sparse semantics, where previous models easily confuse different desert images after feature extraction. However, the SIPN model successfully learns the more fine-grained "green cracks."

L. Sparse Image Experiment

We utilized RemoteCLIP [55] to analyze the correlation between images and text and selected the bottom 20% of image–text pairs with the lowest correlation to create a new sparse image dataset for further comparative experiments. The results are shown in Table XI.

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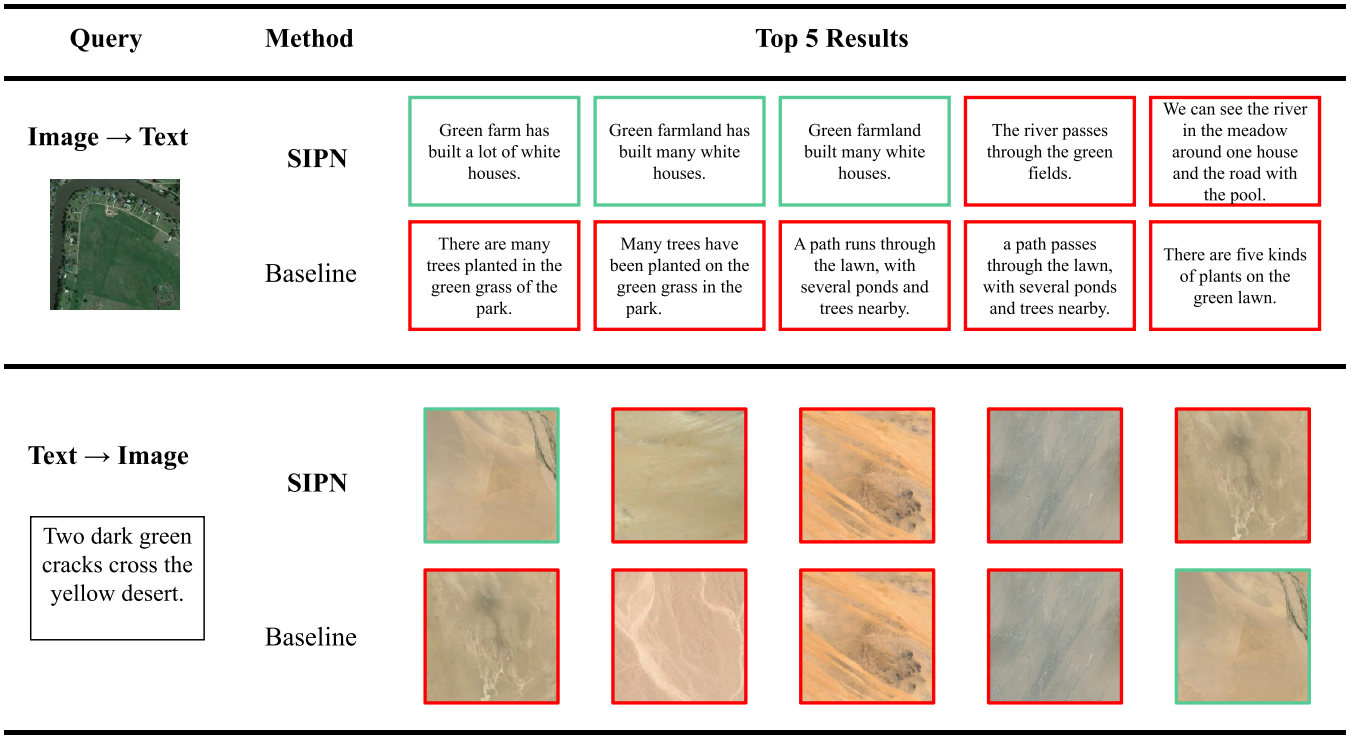


Fig. 6. Visualization of retrieval results. We, respectively, show SIPN and baseline retrieval results on the RSITMD test set when performing image–text and text–image retrieval tasks. In the top-5 results, those surrounded by green boxes are ground truth, and those surrounded by red boxes are wrong results.

TABLE XI
COMPARATIVE ANALYSIS OF SPARSE IMAGES EXTRACTED FROM RSICD AND RSITMD

Method	RSICD							RSITMD						
	Sentence Retrieval			Image Retrieval			mR	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
GALR	1.33	12.19	54.33	2.82	13.11	33.98	19.63	10.12	24.58	38.90	7.98	28.79	44.12	25.75
EnVLF	4.15	23.12	38.89	6.79	24.15	39.12	22.70	14.11	29.89	44.12	11.10	41.11	56.78	32.85
SPIN	6.34	25.17	40.25	7.45	27.12	44.83	25.19	16.84	39.45	53.62	14.92	45.13	62.78	38.79

V. CONCLUSION

This article proposes a novel dynamic framework for capturing various types of information in RS image–text data. We establish a dynamic network that utilizes different training paths to achieve specialized training at the instance level. Additionally, we design a policy network, PAPM, to control the layering process of the dynamic network, which is represented by a random process. To further enhance the model's ability to select and fuse features, we introduce the MLAM to align remote sensing images and texts at different levels. Finally, we validate the effectiveness of this method on multiple datasets using a combination of contrast loss and triplet loss.

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